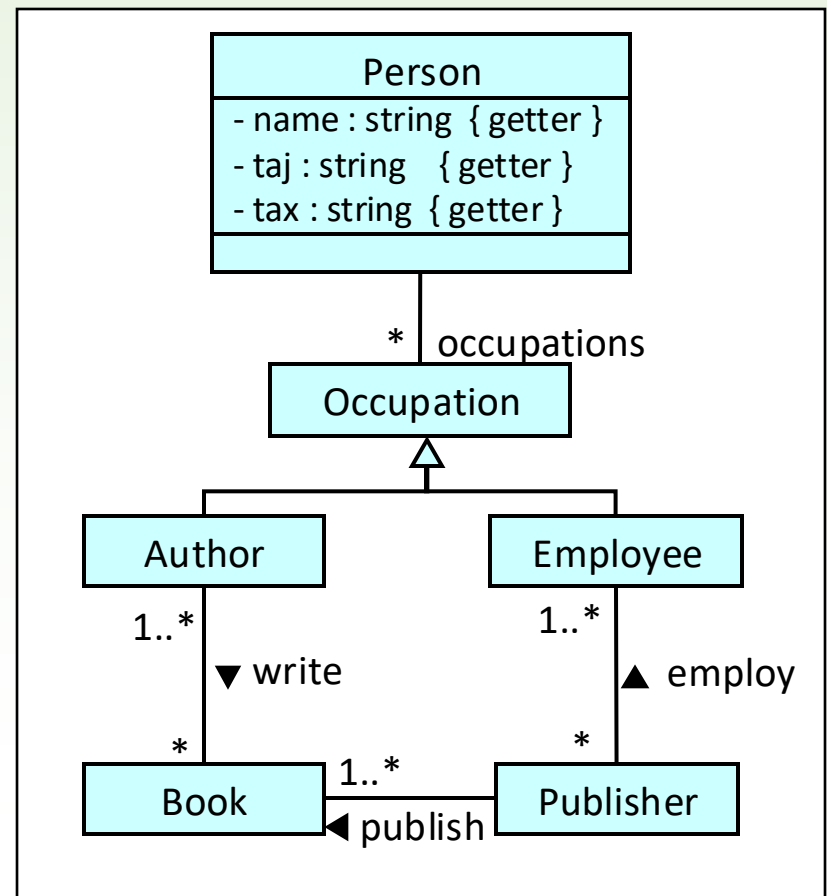
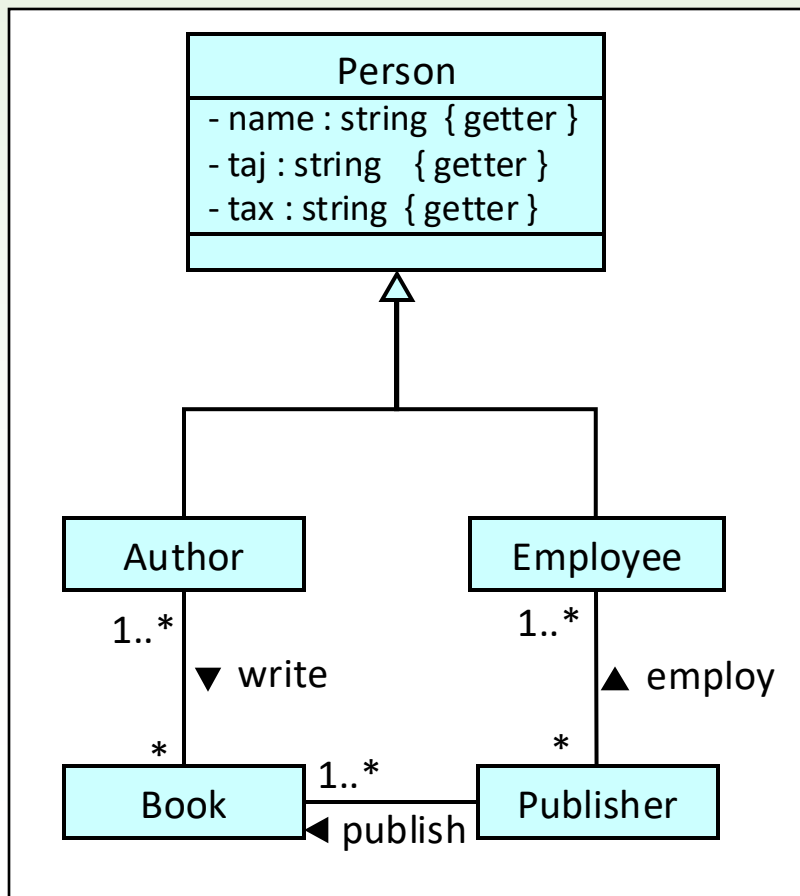


Book publishing

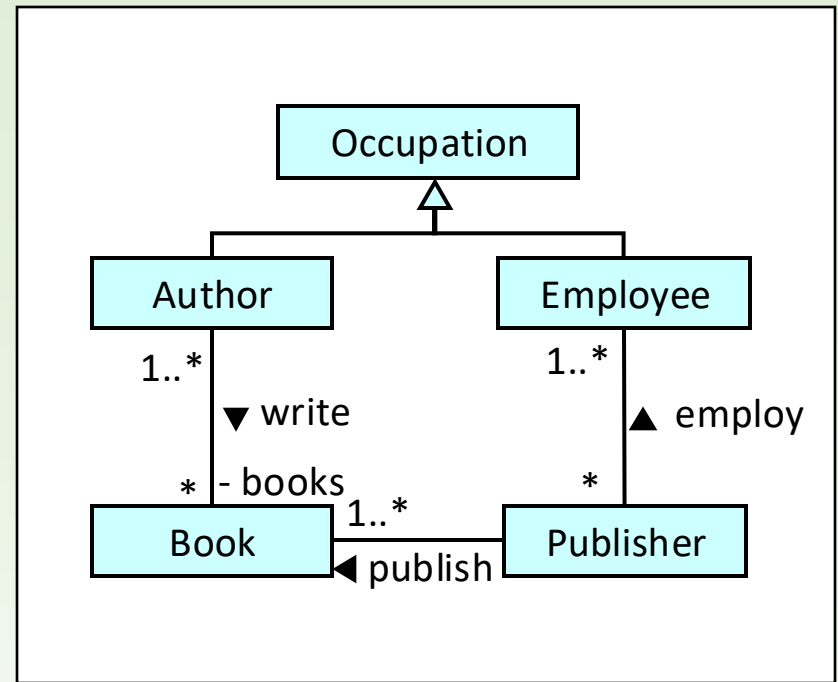
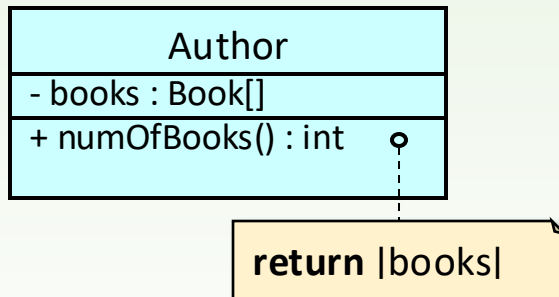
Books are written by at least one author and published by exactly one publisher. A publisher publishes at least one book. A publisher employs at least one employee. Both employees and authors have a name, social security number (TAJ card number), and tax identification number or code. It should be possible for someone to have multiple occupations at the same time.



Book publishing

Task: How many books has a given author written?

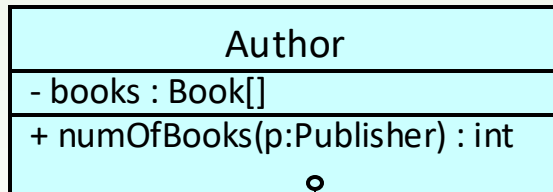
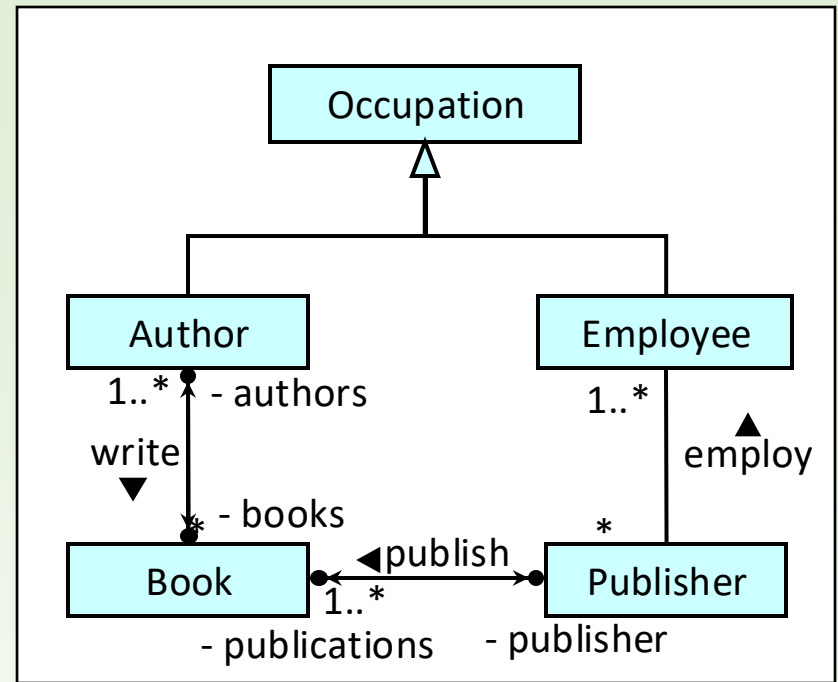
- Which class should numOfBooks() belong to?



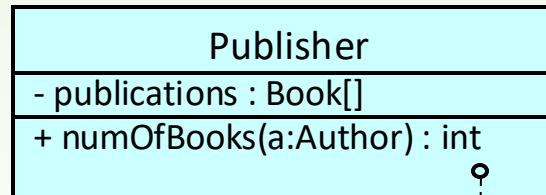
Book publishing

Task: How many books has a given author written for a given publisher?

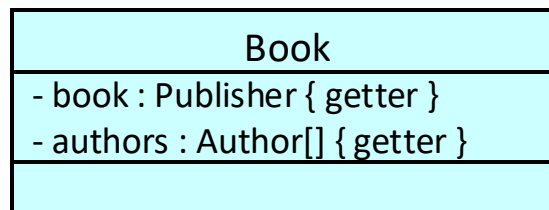
- Which class should numOfBooks() belong to?



return $\sum_{1}^{book \text{ in books } | \text{ book.publisher} = \text{kiad}}$



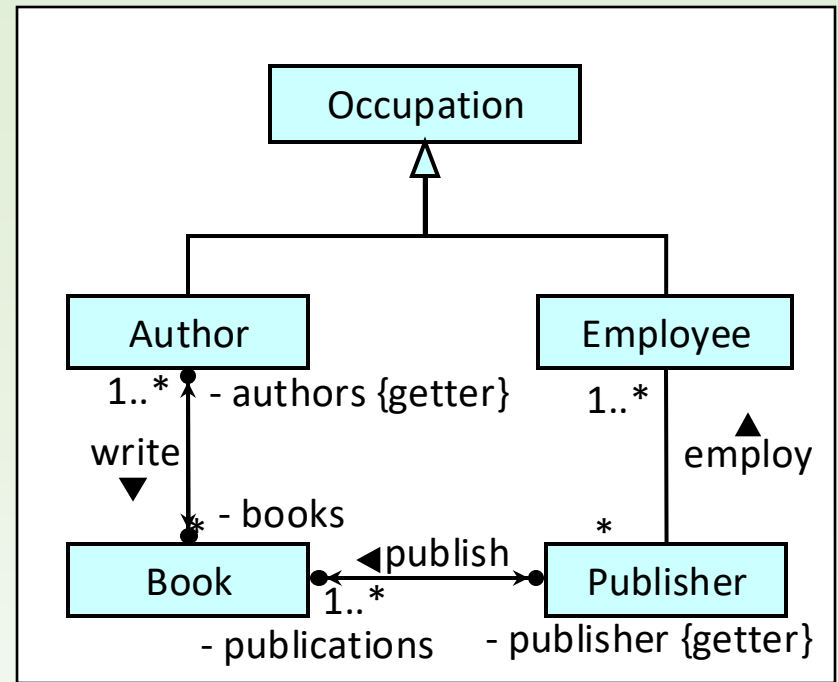
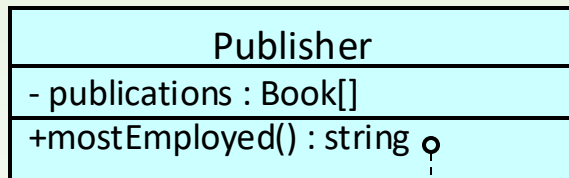
return $\sum_{1}^{book \text{ in publications } | \text{ author in book.authors } | \text{ author} = a}$



Book publishing

Task: Who is the most frequently employed author by a publisher?

- Which class should `mostEmployed()` belong to?



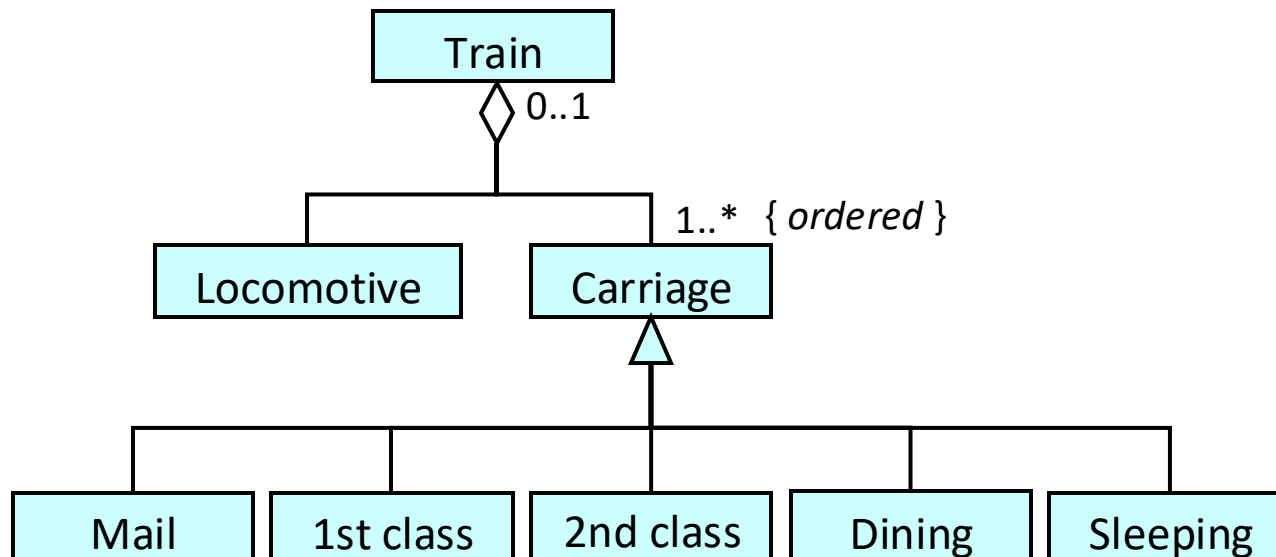
```
(max, maxAuthor) = MAXbook in publications and author in book.authors author.numOfBooks(this)
return maxAuthor
```

Could we come up with a more efficient solution?

Yes, by establishing a direct relation between Author and Publisher.

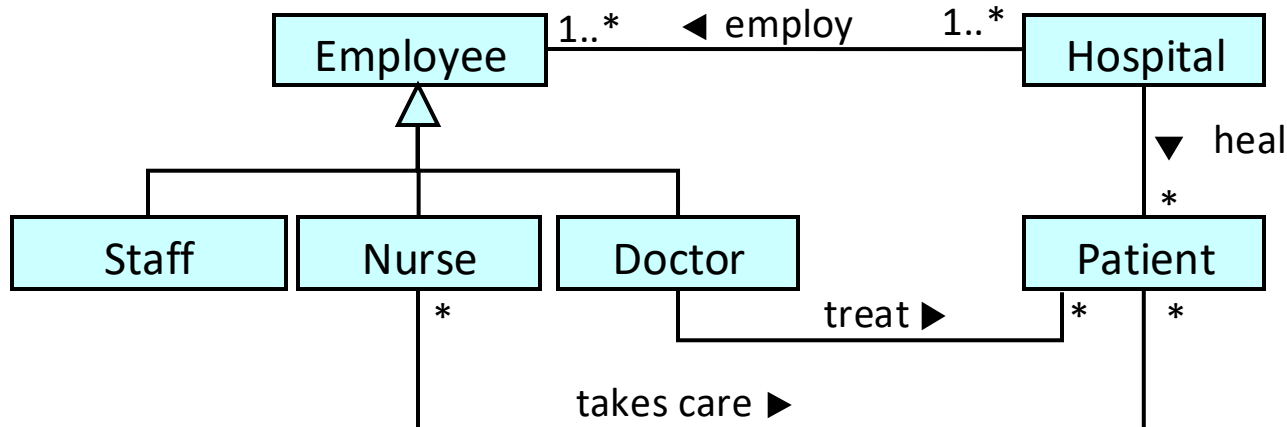
Train

A train consists of a locomotive and at least one carriage. The carriages are connected to the locomotive in a given order. A train can be made up of different types of carriages. Possible types are: first class, second class, mail, dining, sleeping. A locomotive or carriage can only belong to one train at a time.



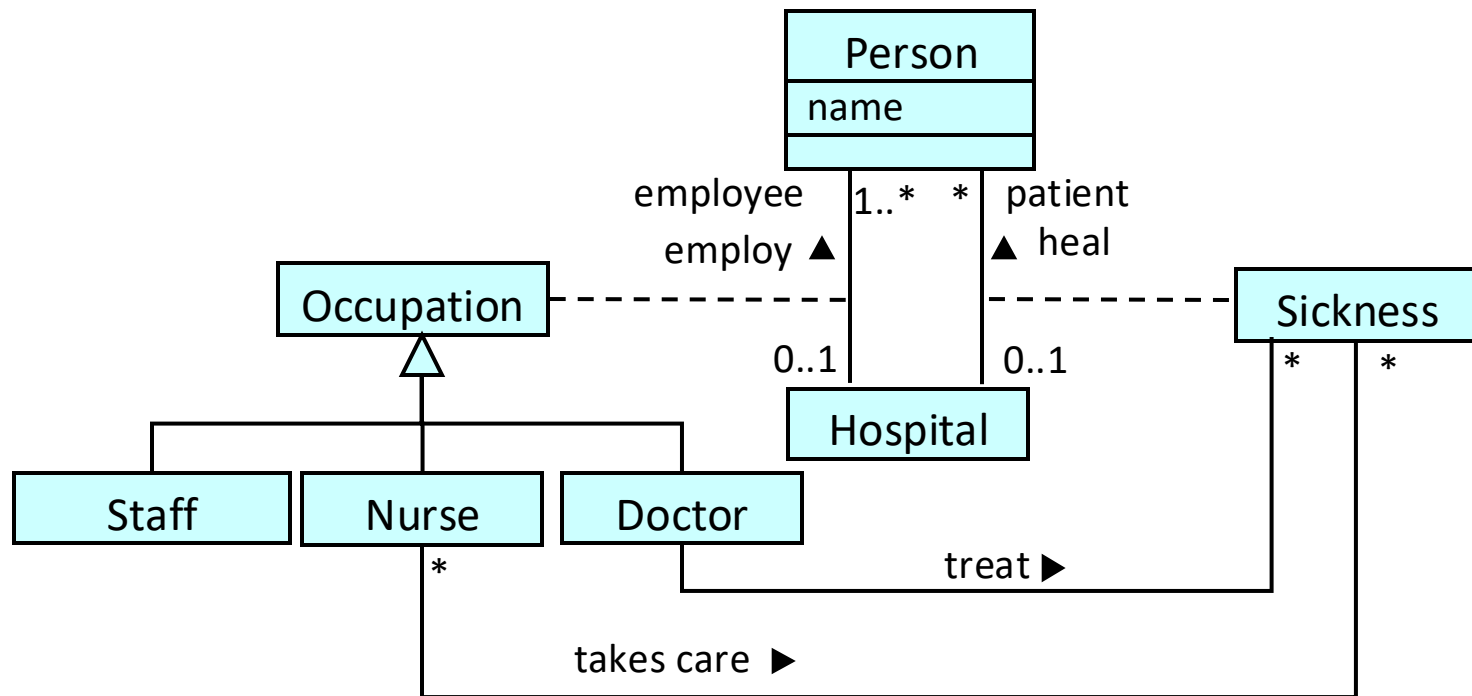
Hospital

In the hospital, there is at least one employee, who can be a doctor, a nurse, or part of the administrative staff. The hospital has patients who are treated by doctors and cared for by nurses. A patient is treated by exactly one doctor and can be cared for by any number of nurses. A doctor can treat any number of patients, and a nurse can care for any number of patients.



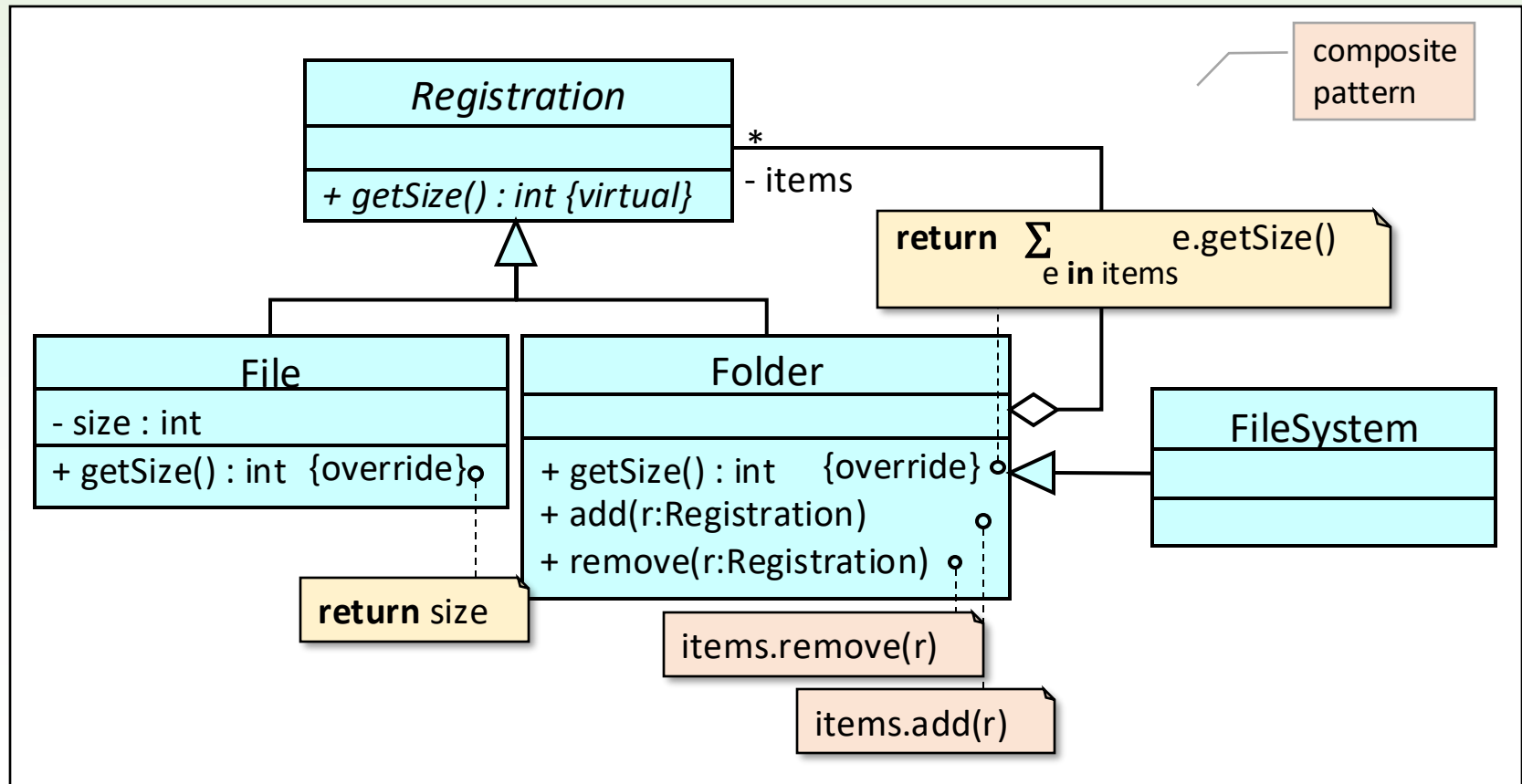
Hospital

Let's also consider that an employee can be sick, and the illness does not end their employment. Modify the class diagram accordingly.



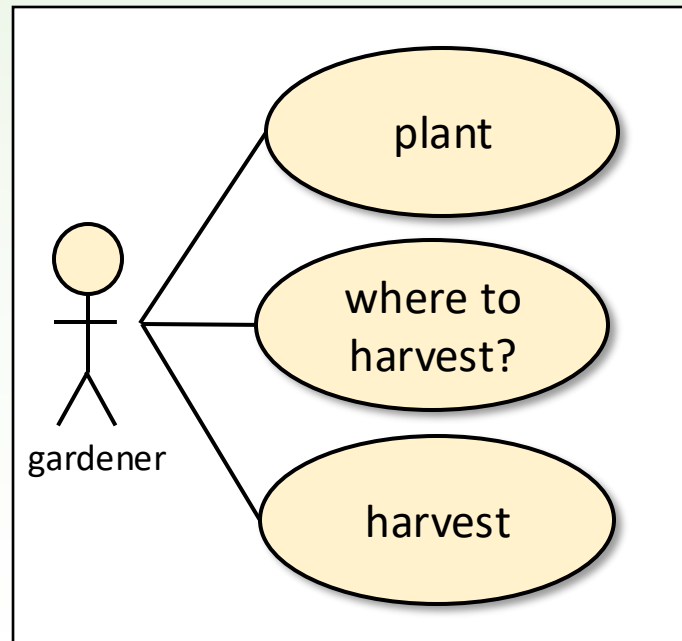
File system

In a computer file system, files are organized into folders. Files can belong directly to the file system or can be located within one of its folders. Each folder can contain any number of files or subfolders. For a file, the occupied storage space is known; the storage space of a folder is the sum of the storage spaces of the files and subfolders it contains. How much storage space does a given folder occupy?



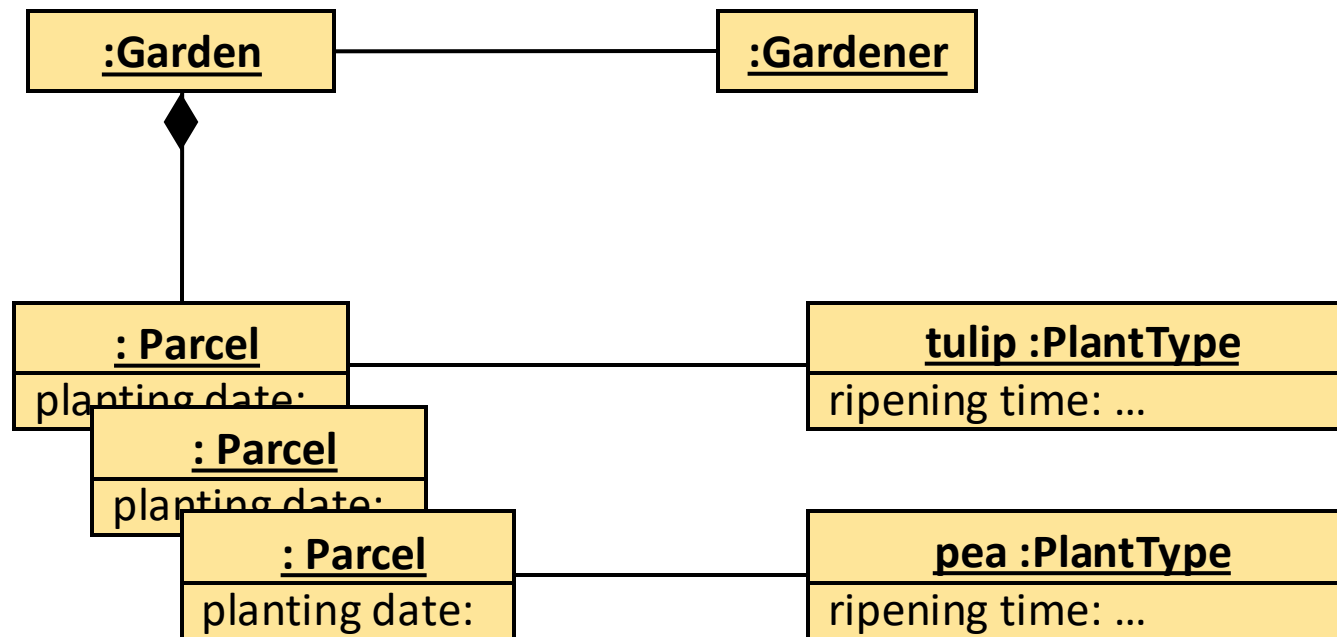
Gardening

A garden is taken care of by a gardener. The garden consists of parcels, and each parcel can be planted with one type of plant. The time of planting is stored (in months). The plants can be vegetables, such as potatoes, peas, or onions; or flowers, such as tulips, carnations, or roses. The ripening time of the plants is also known (in months). The gardener should list the parcels where vegetables will ripen in a given month. (In other words: which parcels are harvestable?)



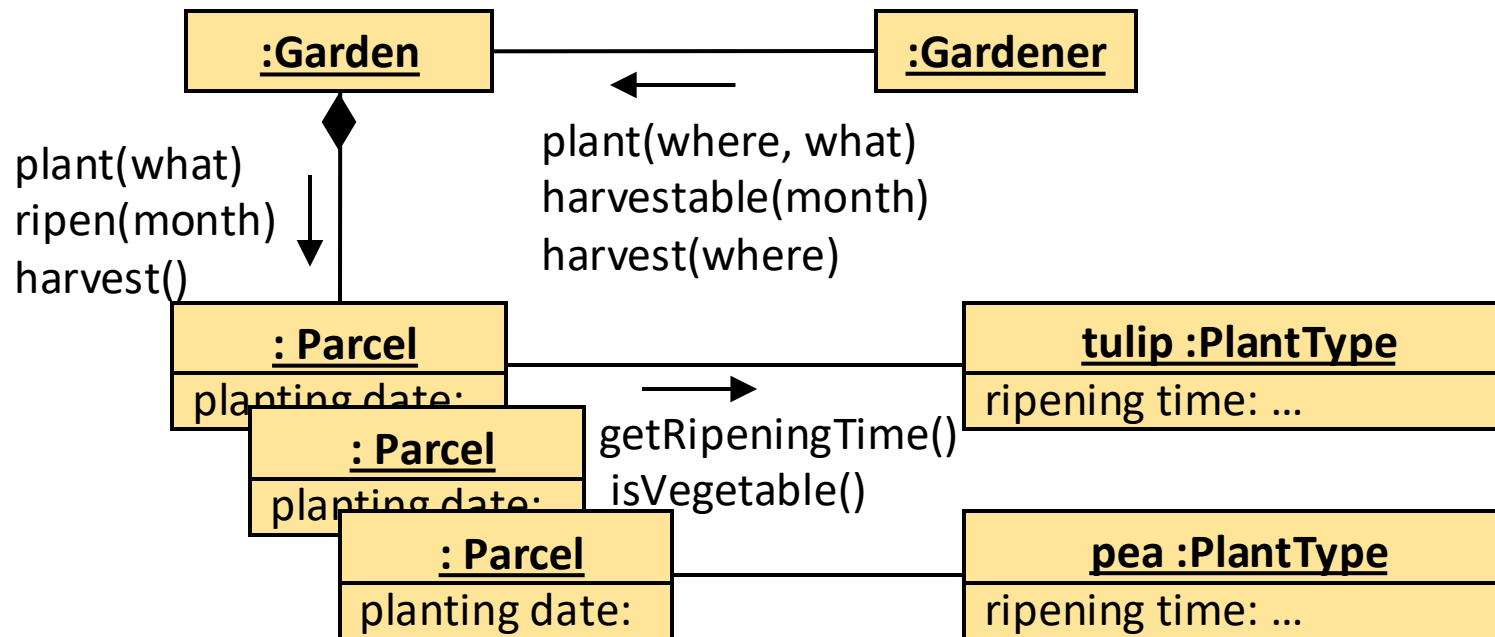
Gardening

A **garden** is taken care of by a **gardener**. The garden consists of **parcels**, and each parcel can be planted with one type of plant. The time of planting is stored (in months). The plants can be vegetables, such as potatoes, peas, or onions; or flowers, such as tulips, carnations, or roses. The ripening time of the plants is also known (in months). The gardener should list the parcels where vegetables will ripen in a given month. (In other words: which parcels are harvestable?)

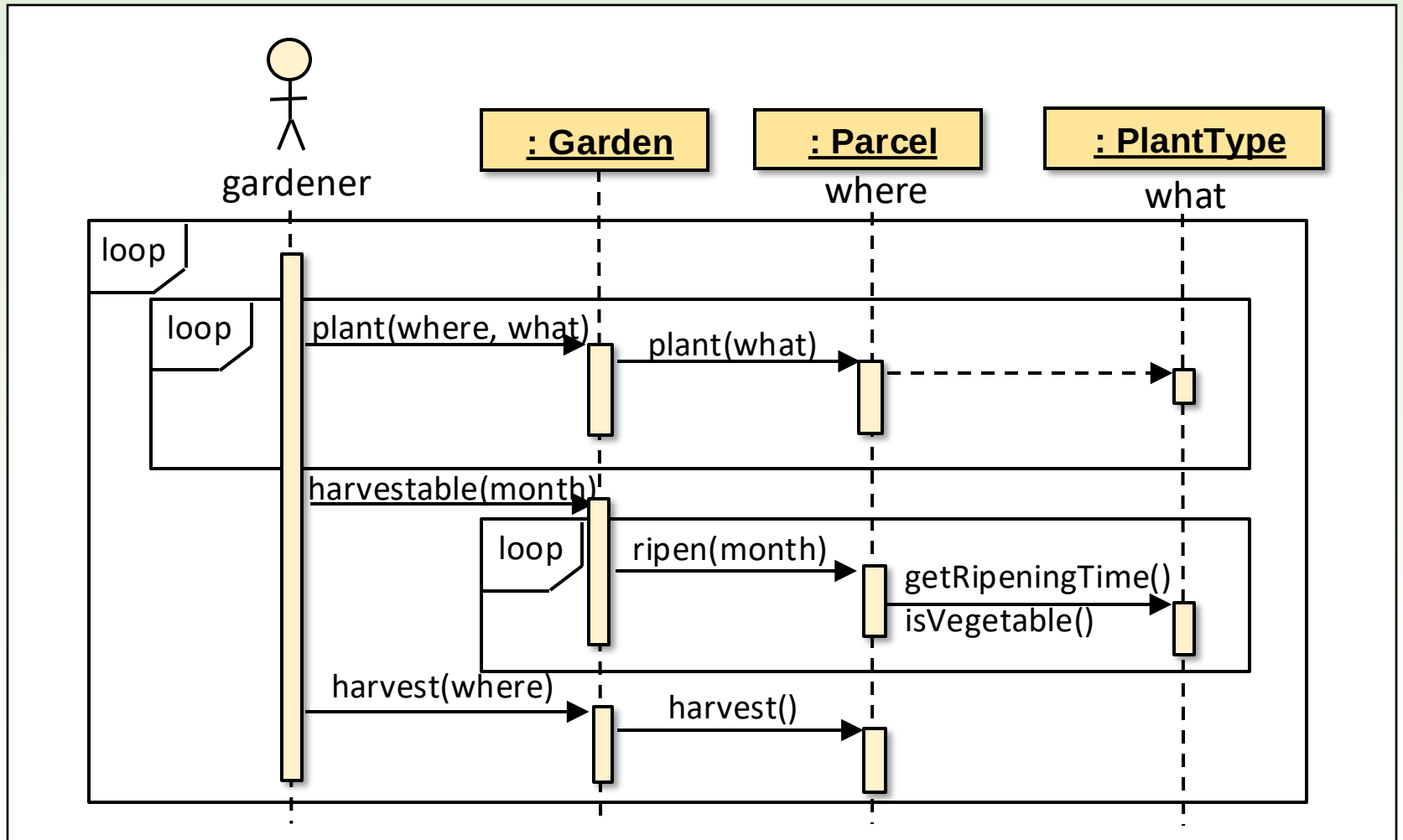


Gardening

A garden is taken care of by a gardener. The garden consists of parcels, and each parcel can be **planted** with one type of plant. The time of planting is stored (in months). The plants can be vegetables, such as potatoes, peas, or onions; or flowers, such as tulips, carnations, or roses. The ripening time of the plants is also known (in months). The gardener should list the parcels where vegetables will **ripen** in a given month. (In other words: which parcels are **harvestable**?)



Gardening



Gardening

